

PRACTICAL WEEK-9

1 . Implement a file copy utility using low-level Linux file I/O system calls (`open`, `read`, `write`, `lseek`, `close`) and compare its performance with standard library functions (`fopen`, `fread`, `fwrite`).

Aim : To implement file copying using low-level system calls and standard C library functions, then measure and compare their execution performance.

Program :

```
GNU nano 7.2
Hello from input file.
This is Linux I/O redirection.
OSSP Practical 9. |
```

```
GNU nano 7.2
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <stdlib.h>

#define BUFFER_SIZE 4096

int main()
{
    int src, dest;
    char buffer[BUFFER_SIZE];
    ssize_t bytes_read, bytes_written;

    src = open("source.txt", O_RDONLY);

    if (src == -1)
    {
        perror("Error opening source file");
        return 1;
    }

    dest = open("lowlevel_copy.txt",
                O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (dest == -1)
    {
        perror("Error creating destination file");
        close(src);
        return 1;
    }

    while ((bytes_read = read(src, buffer, BUFFER_SIZE)) > 0)
    {
        bytes_written = write(dest, buffer, bytes_read);

        if (bytes_written != bytes_read)
        {
            perror("Write error");
            close(src);
            close(dest);
            return 1;
        }
    }

    if (bytes_read == -1)
    {
        perror("Read error");
    }

    lseek(src, 0, SEEK_SET);

    close(src);
    close(dest);

    printf("File copied successfully using low-level system calls.\n");

    return 0;
}
```

GNU nano 7.2

```
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>

int main()
{
    int input_fd;
    int output_fd;

    input_fd = open("input.txt", O_RDONLY);

    if (input_fd == -1)
    {
        perror("Error opening input file");
        return 1;
    }

    output_fd = open("output.txt",
                    O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (output_fd == -1)
    {
        perror("Error opening output file");
        close(input_fd);
        return 1;
    }

    if (dup2(input_fd, STDIN_FILENO) == -1)
    {
        perror("dup2 stdin");
        return 1;
    }

    if (dup2(output_fd, STDOUT_FILENO) == -1)
    {
        perror("dup2 stdout");
        return 1;
    }

    close(input_fd);
    close(output_fd);

    char buffer[100];

    while (fgets(buffer, sizeof(buffer), stdin) != NULL)
    {
        printf("%s", buffer);
    }

    return 0;
}
```

GNU nano 7.2

```
#include <stdio.h>
#include <stdlib.h>

#define BUFFER_SIZE 4096

int main()
{
    FILE *src, *dest;
    char buffer[BUFFER_SIZE];
    size_t bytes_read;

    src = fopen("source.txt", "rb");

    if (src == NULL)
    {
        perror("Error opening source file");
        return 1;
    }

    dest = fopen("library_copy.txt", "wb");

    if (dest == NULL)
    {
        perror("Error creating destination file");
        fclose(src);
        return 1;
    }

    while ((bytes_read = fread(buffer, 1, BUFFER_SIZE, src)) > 0)
    {
        if (fwrite(buffer, 1, bytes_read, dest) != bytes_read)
        {
            perror("Write error");
            fclose(src);
            fclose(dest);
            return 1;
        }
    }

    fclose(src);
    fclose(dest);

    printf("File copied successfully using standard library functions.\n");

    return 0;
}
```

Output :

```
root@keerthana: ~/OSSP
root@keerthana:~/OSSP# cd ~/OSSP
root@keerthana:~/OSSP# mkdir -p ~/OSSP
root@keerthana:~/OSSP# cd ~/OSSP
root@keerthana:~/OSSP# nano source.txt
root@keerthana:~/OSSP# nano lowlevel_copy.c
root@keerthana:~/OSSP# gcc -Wall -Wextra lowlevel_copy.c -o lowlevel_copy
root@keerthana:~/OSSP# ./lowlevel_copy
File copied successfully using low-level system calls.
root@keerthana:~/OSSP# cat lowlevel_copy.txt
Hello from input file.
This is Linux I/O redirection.
OSSP Practical 9.
root@keerthana:~/OSSP# nano library_copy.c
root@keerthana:~/OSSP# gcc -Wall -Wextra library_copy.c -o library_copy
root@keerthana:~/OSSP# ./library_copy
File copied successfully using standard library functions.
root@keerthana:~/OSSP# cat library_copy.txt
Hello from input file.
This is Linux I/O redirection.
OSSP Practical 9.
root@keerthana:~/OSSP# cmp source.txt lowlevel_copy.txt
root@keerthana:~/OSSP# cmp source.txt library_copy.txt
root@keerthana:~/OSSP# time ./lowlevel_copy
File copied successfully using low-level system calls.

real    0m0.003s
user    0m0.003s
sys      0m0.001s
root@keerthana:~/OSSP# time ./library_copy
File copied successfully using standard library functions.

real    0m0.003s
user    0m0.002s
sys      0m0.001s
root@keerthana:~/OSSP# |
```

2 . Develop a program that redirects standard input and output using file descriptors and `dup2 ()` , demonstrating how shells implement I/O redirection.

Aim : To demonstrate standard input and output redirection using file descriptors and the `dup2 ()` system call.

Program :

```
GNU nano 7.2
Hello from input file.
This is Linux I/O redirection.
OSSP Practical 9.
```

```
GNU nano 7.2
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>

int main()
{
    int input_fd;
    int output_fd;

    input_fd = open("input.txt", O_RDONLY);
    if (input_fd == -1)
    {
        perror("Error opening input file");
        return 1;
    }

    output_fd = open("output.txt",
                    O_WRONLY | O_CREAT | O_TRUNC, 0644);
    if (output_fd == -1)
    {
        perror("Error opening output file");
        close(input_fd);
        return 1;
    }

    if (dup2(input_fd, STDIN_FILENO) == -1)
    {
        perror("dup2 stdin");
        return 1;
    }

    if (dup2(output_fd, STDOUT_FILENO) == -1)
    {
        perror("dup2 stdout");
        return 1;
    }

    close(input_fd);
    close(output_fd);

    char buffer[100];
    while (fgets(buffer, sizeof(buffer), stdin) != NULL)
    {
        printf("%s", buffer);
    }

    return 0;
}
```

Output :

```
root@keerthana:~/OSSP# cd ~/OSSP
root@keerthana:~/OSSP# nano input.txt
root@keerthana:~/OSSP# cat input.txt
Hello from input file.
This is Linux I/O redirection.
OSSP Practical 9.
root@keerthana:~/OSSP# ls -l input.txt
-rw-r--r-- 1 root root 72 Sep 27 15:12 input.txt
root@keerthana:~/OSSP# nano io_redirect.c
root@keerthana:~/OSSP# gcc -Wall -Wextra io_redirect.c -o io_redirect
root@keerthana:~/OSSP# ./io_redirect
root@keerthana:~/OSSP# dup2(input_fd, STDIN_FILENO);
-bash: syntax error near unexpected token `input_fd,'
root@keerthana:~/OSSP# input.txt
input.txt: command not found
root@keerthana:~/OSSP# cat output.txt
Hello from input file.
This is Linux I/O redirection.
OSSP Practical 9.
root@keerthana:~/OSSP# cat input.txt
Hello from input file.
This is Linux I/O redirection.
OSSP Practical 9.
root@keerthana:~/OSSP# diff input.txt output.txt
root@keerthana:~/OSSP# |
```